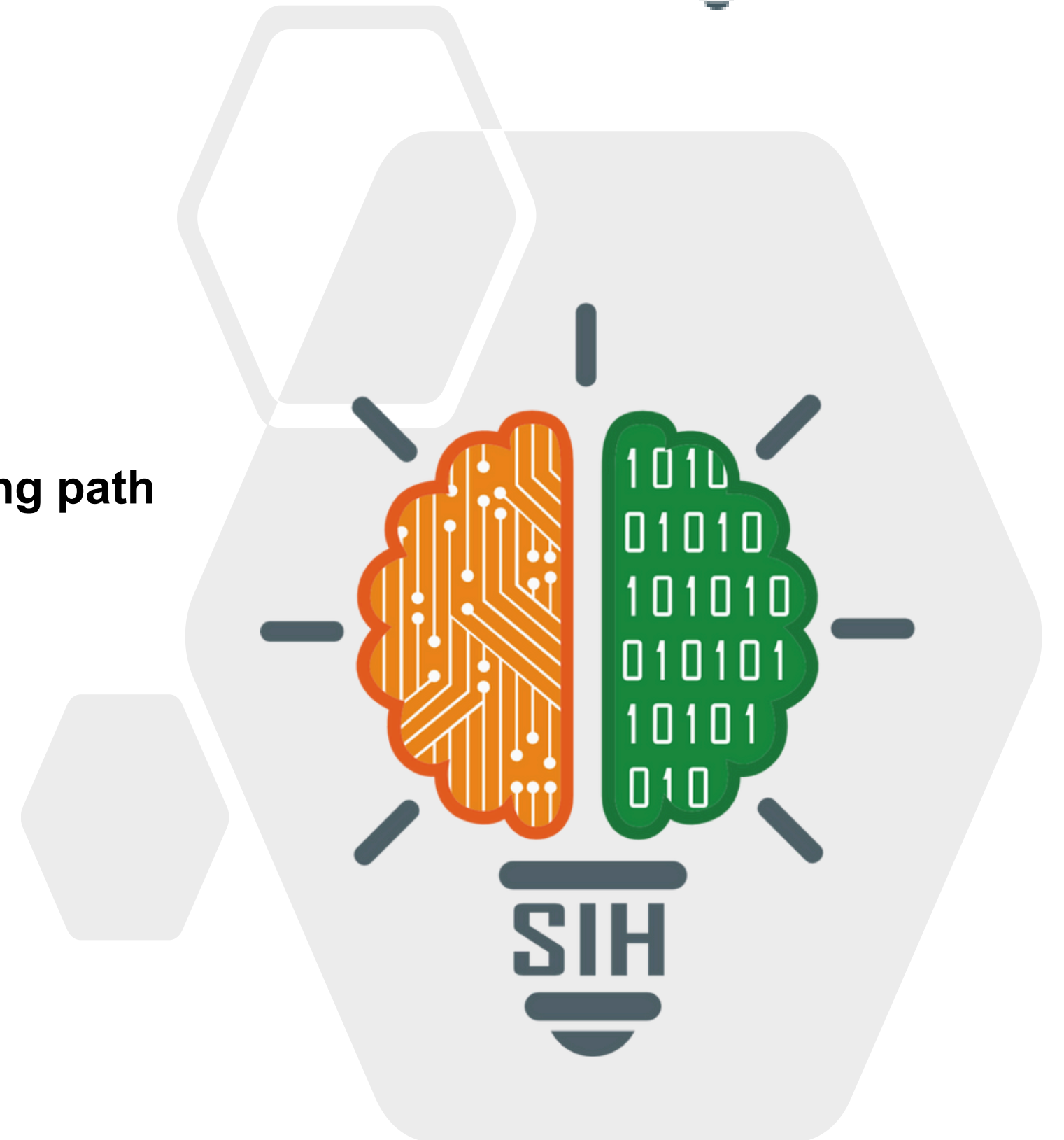


TITLE PAGE

- **Problem Statement ID – SIH25199**
- **Problem Statement Title- AI Powered Personalized learning path generator**
- **Theme- Education & Career Development**
- **PS Category- Smart Education**
- **Software Team ID- 53748**
- **Team Name - Cr8iveX**



AI - POWERED PERSONALISED LEARNING PATH GENERATOR

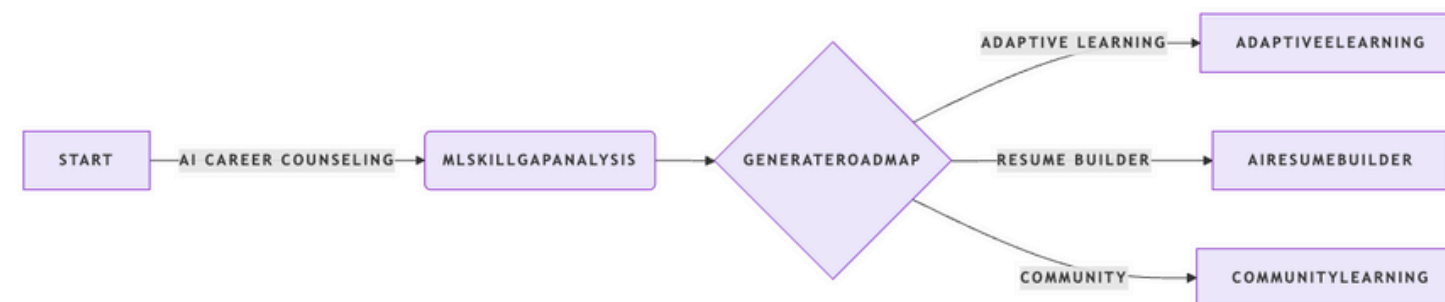
Proposed Solution (Cr8iveX)

An AI & ML-driven platform for personalized student career Path creator .

- Uses AI to create custom learning paths based on:
 - Student's current skill level
 - Learning style (visual, textual, interactive)
 - Interests and career goals
 - Performance in previous assessments
- Dynamically adjusts the path as the student progresses.
- Provides actionable recommendations, resources, quizzes, and timelines.

Key Features:

- Skill gap analysis
- Adaptive learning plan
- Progress tracking and feedback
- Integration with existing learning platforms

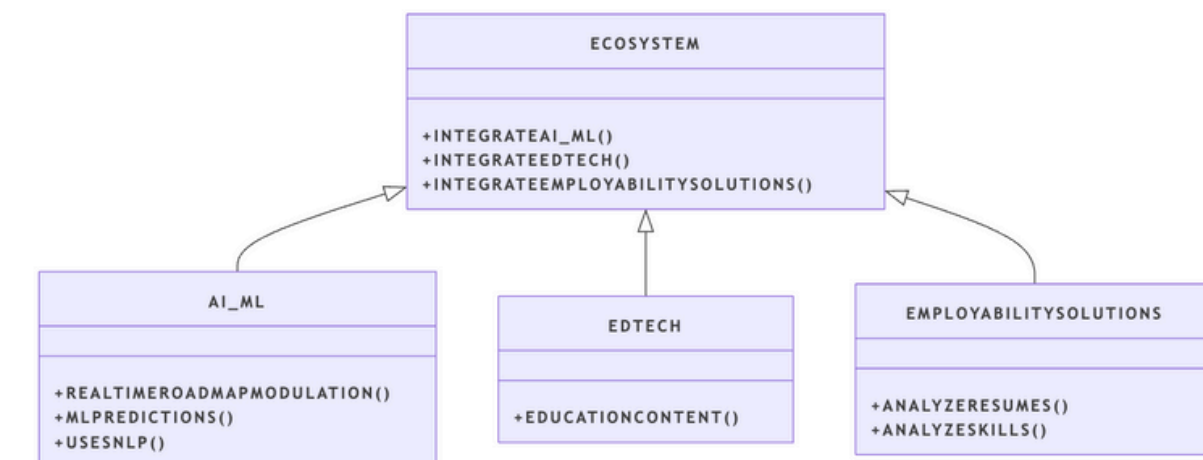


Innovation & Uniqueness:

Integrates AI/ML + EdTech + Employability Solutions in one ecosystem.

Real-time roadmap modulation using ML predictions.

Uses NLP for analyzing resumes & skills.



Technologies Used:

- Frontend – React.js, Tailwind CSS
- Backend – Node.js, Express Database –
- MongoDB, Firebase AI/ML – Python,
- TensorFlow, Scikit-learn NLP – OpenAI
- APIs, Hugging Face Transformers (for resume/skill parsing) Cloud Infra –
- AWS/GCP with scalable microservices



Methodology :

Step 1 – Data Collection

- Student details: academics, skills, interests, and goals.
- Industry datasets: skill benchmarks, job role requirements.

Step 2 – AI/ML Processing

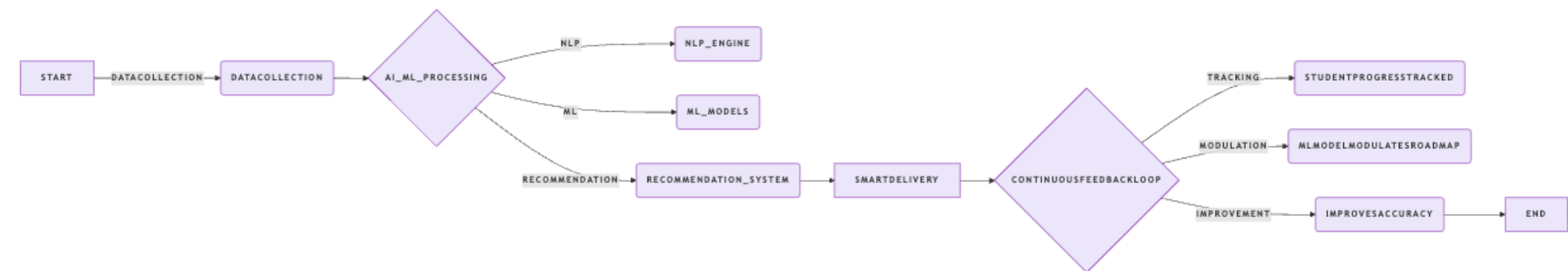
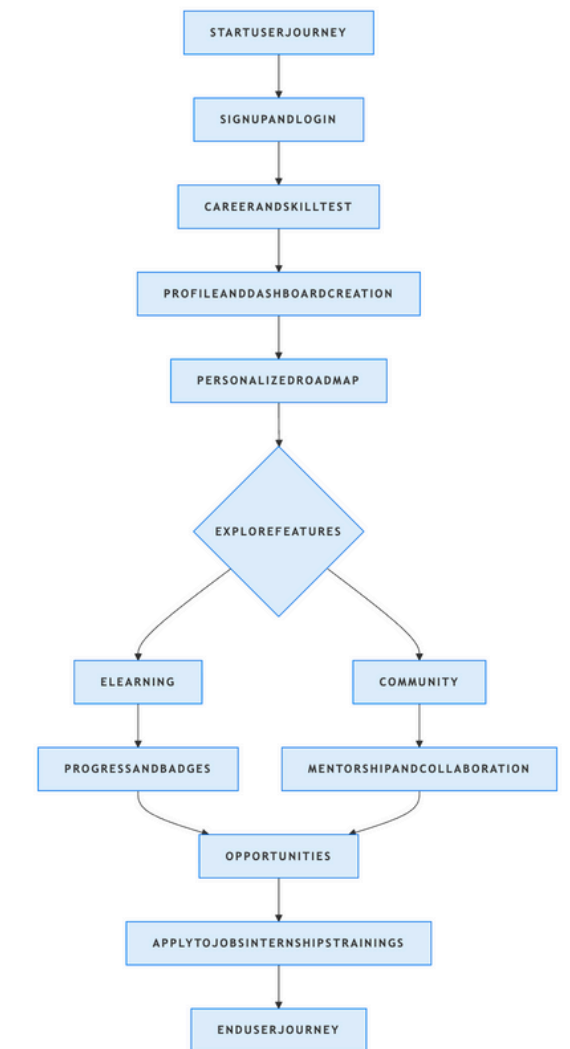
- NLP Engine → Analyzes student input & resumes.
- ML Models → Detect skill gaps, predict career fit.
- Recommendation System → Generates personalized learning roadmap.

Step 3 – Smart Delivery

- Adaptive e-learning modules (tech + soft skills).
- AI-powered resume builder & interview practice.
- Gamification + mentor network for engagement.

Step 4 – Continuous Feedback Loop

- Student progress tracked in real time.
- ML model modulates roadmap dynamically.
- Improves accuracy with every user interaction.



Feasibility:

- Uses existing AI/ML libraries & scalable cloud infra.
- Works as both B2C (students) & B2B (colleges/ed-institutions).

Challenges:

- Data privacy & personalization accuracy.
- Onboarding large student base.

Strategies:

- Secure data encryption (GDPR compliance).
- Partnerships with colleges for adoption.
- Continuous AI training with real user data

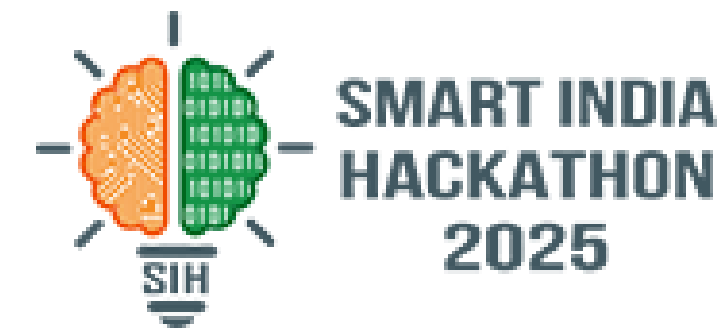
Impact:

- Reduces unemployability by bridging education–industry gap.
- Increases student confidence & clarity in career.
- Helps institutions improve placement statistics.

Benefits:

- Social – Empowered youth, reduced frustration.
- Economic–Job-ready graduates, boost to employment.
- Technological– India-first AI/ML solution for employability.

RESEARCH AND REFERENCES



- NASSCOM Report (2023) – 85% of Indian graduates considered unemployable due to lack of skills.
- World Economic Forum – Future of Jobs Report (2023) – AI, data science, and green jobs will dominate the next decade; highlights urgent need for upskilling & reskilling.
- McKinsey Global Survey on AI in Education (2022) – AI-powered learning boosts student engagement by 30–40%.
- UNESCO Report (2023) – Personalized learning pathways are critical to bridging global skill gaps.
- LinkedIn Workplace Learning Report (2024) – Soft skills + adaptability are top hiring priorities, not just technical knowledge.
- IEEE Research Papers:
 - “Machine Learning in Skill Gap Analysis and Career Counseling” – demonstrates ML’s role in employability prediction.
 - “NLP-driven Resume Parsing for Automated Career Guidance” – AI can match resumes with job market trends.
- Market Research Future (MRFR) Report (2024) – AI in EdTech market projected to reach \$20B+ by 2027.
- Harvard Business Review (HBR) – Data-driven career counseling improves placement outcomes by 25%.